INCH POUND
MIL-PRF-3098/10H
05 March 2010
SUPERSEDING
MIL-PRF-3098/10G
19 December 2003

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR28/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental: controlled; series resonance.

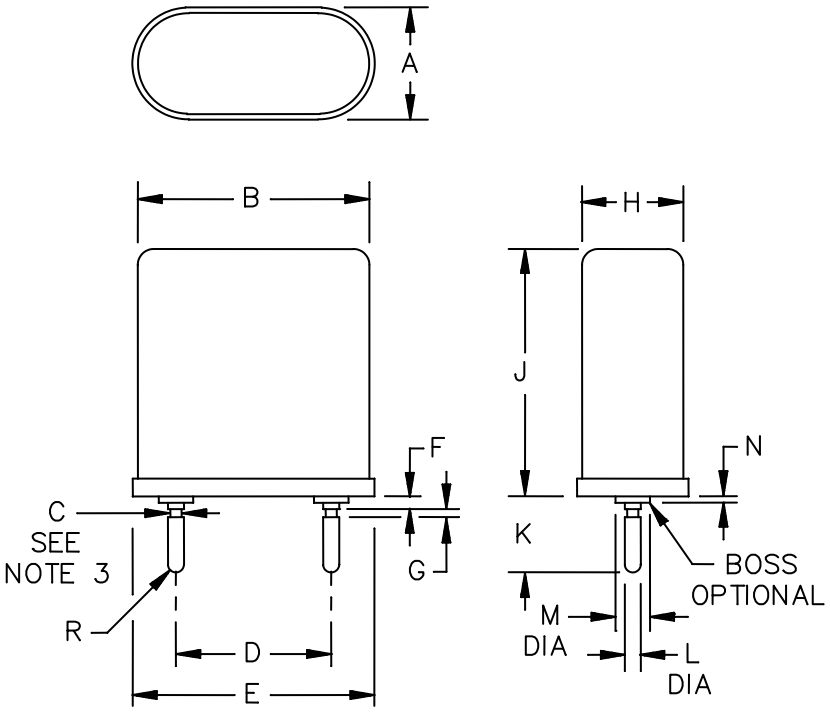


FIGURE 1. Crystal unit - CR28/U.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:18Z
Check the source to verify that this is the current version before use.

Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.352	---	8.94
B	---	.725	---	18.42
C	.030	.037	0.76	0.94
D	.478	.494	12.14	12.55
E	---	.757	---	19.23
F	.030	.040	0.76	1.02
G	.015	.025	0.38	0.64
H	---	.317	---	8.05
J	---	.775	---	19.69
K	.223	.248	5.66	6.30
L	.048	.052	1.22	1.32
M	.075	.141	1.91	3.58
N	.015	.025	0.38	0.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR28/U - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:18Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20 MHz, inclusive.

Capacitance, shunt: 7 pF, maximum.

Frequency tolerance:

Operating temperature range: ±20 parts per million (ppm).

Room temperature: ±70 ppm.

Frequency stability: ±5 ppm.

Equivalent resistance: See table I.

Mode of oscillation: Fundamental.

Reference temperature: +75°C ±1°C.

Temperature ranges:

Operable: -55°C to +70°C, and +80°C to +90°C, inclusive

Operating (controlled): +70°C to +80°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Shock (specified pulse):

Frequency change permitted: ± 5 ppm.

Equivalent resistance change permitted: Below 2MHz: ±15 percent
2.0 MHz and above: ±10 percent

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ± 5 ppm.

Equivalent resistance change permitted: Below 2MHz: ±15 percent
2.0 MHz and above: ±10 percent

Thermal shock:

Frequency change permitted: ± 5 ppm.

Equivalent resistance change permitted: Below 2MHz: ±15 percent
2.0 MHz and above: ±10 percent

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:18Z
Check the source to verify that this is the current version before use.

Aging:

Frequency change permitted: ± 5 ppm.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.80 to 0.85	520
0.85+ to 0.90	480
0.90+ to 1.00	440
1.00+ to 1.12	400
1.12+ to 1.25	380
1.25+ to 1.37	340
1.37+ to 1.50	300
1.50+ to 1.62	270
1.62+ to 1.75	250
1.75+ to 1.87	220
1.87+ to 2.00	180
2.00+ to 2.12	160
2.12+ to 2.25	150
2.25+ to 2.60	120
2.60+ to 3.00	90
3.00+ to 3.40	70
3.40+ to 3.75	52
3.75+ to 4.00	45
4.00+ to 5.00	37
5.00+ to 7.00	25
7.00+ to 10.00	20
10.00+ to 15.00	18
15.00+ to 20.00	15

Part or Identifying Number (PIN): CR28/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Referenced documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:18Z
Check the source to verify that this is the current version before use.

Custodians:
Army – CR
Navy - EC
Air force - 99
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2009-027)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:18Z
Check the source to verify that this is the current version before use.